

JEE + NEET

MASTER PRACTICE WORKBOOK

■ Physics

■ Chemistry

■ Mathematics

■ Biology

123
Physics Qs

120
Chemistry Qs

133
Maths Qs

143
Biology Qs

■ Total Questions: 519

Covers full JEE Mains & Advanced + NEET syllabus

MCQ • Short Answer • Numerical Problems

PYQ-inspired + AI-generated practice questions

Answer key provided at the end of each subject section

Prepared with AI assistance | For self-practice only

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■ BIOLOGY

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■ PHYSICS

■ Mechanics

Q.1

MCQ

A ball is thrown vertically upward with a velocity of 20 m/s. The maximum height reached is ($g = 10 \text{ m/s}^2$):

- (A) 10 m
- (B) 20 m
- (C) 40 m
- (D) 5 m

Q.2

MCQ

A body of mass 2 kg is acted upon by two perpendicular forces of 3 N and 4 N. The acceleration of the body is:

- (A) 2.5 m/s^2
- (B) 3.5 m/s^2
- (C) 1.5 m/s^2
- (D) 5 m/s^2

Q.3

MCQ

A projectile is fired at 45° with initial speed u . The range of the projectile is:

- (A) u^2/g
- (B) $u^2/2g$
- (C) $2u^2/g$
- (D) u/g

Q.4

MCQ

The kinetic energy of a body of mass 4 kg moving with velocity 5 m/s is:

- (A) 20 J
- (B) 40 J
- (C) 50 J
- (D) 100 J

Q.5

MCQ

Newton's second law of motion states that force equals:

- (A) mass $\times$ velocity
- (B) mass $\times$ acceleration
- (C) mass $\times$ displacement
- (D) velocity $\times$ time

Q.6

MCQ

A car accelerates uniformly from rest to 72 km/h in 10 s. The acceleration is:

- (A) 2 m/s^2
- (B) 7.2 m/s^2
- (C) 0.2 m/s^2
- (D) 3.6 m/s^2

Q.7

MCQ

The SI unit of angular momentum is:

- (A) $\text{kg}\cdot\text{m/s}$
- (B) $\text{kg}\cdot\text{m}^2/\text{s}$
- (C) $\text{N}\cdot\text{m}$
- (D) $\text{J}\cdot\text{s}^2$

Q.8

MCQ

A stone tied to a string moves in a circle. Which force acts as centripetal force?

- (A) Gravity
- (B) Normal force
- (C) Tension
- (D) Friction

Q.9

MCQ

Work done by a force is zero when the angle between force and displacement is:

- (A) 0°
- (B) 45°
- (C) 90°
- (D) 180°

Q.10

MCQ

The momentum of a 10 kg body moving at 5 m/s is:

- (A) $2 \text{ kg}\cdot\text{m/s}$
- (B) $15 \text{ kg}\cdot\text{m/s}$
- (C) $50 \text{ kg}\cdot\text{m/s}$
- (D) $0.5 \text{ kg}\cdot\text{m/s}$

Q.11

MCQ

A body slides down a frictionless inclined plane of angle 30° . Its acceleration is ($g = 10 \text{ m/s}^2$):

- (A) 5 m/s^2
- (B) 10 m/s^2
- (C) 8.66 m/s^2
- (D) 2.5 m/s^2

Q.12

MCQ

The escape velocity from the surface of the Earth is approximately:

- (A) 8 km/s
- (B) 11.2 km/s
- (C) 16 km/s
- (D) 3 km/s

Q.13

MCQ

A spring of spring constant 200 N/m is compressed by 0.05 m. The potential energy stored is:

- (A) 0.25 J
- (B) 0.5 J
- (C) 5 J
- (D) 10 J

Q.14

MCQ

The time period of a simple pendulum of length 1 m on Earth ($g = 10 \text{ m/s}^2$) is approximately:

- (A) 1 s
- (B) 2 s
- (C) 3.14 s
- (D) 0.5 s

Q.15

MCQ

Conservation of momentum applies when:

- (A) Net external force is zero
- (B) Net torque is zero
- (C) Energy is conserved
- (D) Velocity is constant

Q.16

SHORT

The velocity of an object in uniform circular motion is constant in _____ but changes in _____.

Ans: _____

Q.17

NUMERICAL

A 5 kg object is placed on a frictionless surface. A horizontal force of 15 N acts on it. Find the distance travelled in 4 seconds starting from rest.

Ans: _____

Q.18

SHORT

Distinguish between elastic and inelastic collisions with one example each.

Ans: _____

Q.19

NUMERI
CAL

A bullet of mass 0.02 kg is fired with velocity 200 m/s into a block of 0.98 kg at rest on a frictionless surface. Find the common velocity after the bullet embeds in the block.

Ans: _____

Q.20

SHORT

A body is moving in a straight line. Draw the v-t graph for uniform deceleration and explain what the area under the graph represents.

Ans: _____

■ Waves

Q.21

MCQ

The frequency of a wave is 500 Hz and its wavelength is 0.6 m. The speed of the wave is:

- (A) 300 m/s
- (B) 250 m/s
- (C) 500 m/s
- (D) 600 m/s

Q.22

MCQ

In simple harmonic motion, acceleration is maximum at:

- (A) Mean position
- (B) Extreme position
- (C) Between mean and extreme
- (D) Nowhere

Q.23

MCQ

The phenomenon of resonance occurs when the frequency of driving force equals:

- (A) Zero
- (B) Natural frequency of the system
- (C) Double the natural frequency
- (D) Half the natural frequency

Q.24

MCQ

Beats are produced due to:

- (A) Reflection
- (B) Superposition of waves of slightly different frequencies
- (C) Diffraction
- (D) Polarization

Q.25

SHORT

The Doppler effect relates to change in _____ due to relative motion between source and observer.

Ans: _____

Q.26

MCQ

Which law of thermodynamics introduces the concept of entropy?

- (A) Zeroth law
- (B) First law
- (C) Second law
- (D) Third law

Q.27

MCQ

In an adiabatic process:

- (A) Temperature remains constant
- (B) No heat exchange occurs
- (C) Pressure remains constant
- (D) Volume remains constant

Q.28

MCQ

The efficiency of a Carnot engine operating between 127°C and 27°C is:

- (A) 25%
- (B) 33%
- (C) 50%
- (D) 75%

Q.29

MCQ

The internal energy of an ideal gas depends only on:

- (A) Pressure
- (B) Volume
- (C) Temperature
- (D) Density

Q.30

MCQ

Which process is represented by a vertical line on a P-V diagram?

- (A) Isothermal
- (B) Adiabatic
- (C) Isochoric
- (D) Isobaric

Q.31

MCQ

The root mean square speed of gas molecules is proportional to:

- (A) $\sqrt{T}$
- (B) T
- (C) T^2
- (D) $1/T$

Q.32

SHORT

A gas expands isothermally. What happens to its internal energy?

Ans: _____

Q.33

SHORT

State and explain the first law of thermodynamics with an example.

Ans: _____

■ Electrostatics

Q.34

MCQ

The electric field inside a uniformly charged hollow conducting sphere is:

- (A) Maximum at center
- (B) Zero
- (C) Same as outside
- (D) Infinite

Q.35

MCQ

Coulomb's law gives the force between two point charges. If the distance is doubled, the force becomes:

- (A) 4 times
- (B) Half
- (C) Double
- (D) One-fourth

Q.36

MCQ

The unit of electric potential is:

- (A) Newton
- (B) Joule
- (C) Volt
- (D) Coulomb

Q.37

MCQ

A capacitor of $5\ \mu\text{F}$ is connected to a $10\ \text{V}$ battery. The charge stored is:

- (A) $0.5\ \text{C}$
- (B) $50\ \mu\text{C}$
- (C) $2\ \mu\text{C}$
- (D) $0.02\ \text{C}$

Q.38

MCQ

Electric field lines never intersect because:

- (A) They are parallel
- (B) At any point, the field has a unique direction
- (C) They have the same magnitude
- (D) None of the above

Q.39

MCQ

The dielectric constant of a vacuum is:

- (A) 0
- (B) 1
- (C) Infinity
- (D) 8.85×10^{-12}

Q.40

MCQ

The energy stored in a capacitor of capacitance C charged to voltage V is:

- (A) CV
- (B) CV^2
- (C) $\frac{1}{2}CV^2$
- (D) $2CV^2$

Q.41

NUMERICAL

Calculate the electric field at a distance of 0.3 m from a charge of 5×10^{-12} C in free space.

Ans: _____

■ Current Electricity

Q.42

MCQ

Ohm's law states that current is proportional to voltage at constant:

- (A) Pressure
- (B) Temperature
- (C) Resistance
- (D) Both B and C

Q.43

MCQ

Three resistors of 3Ω each are connected in parallel. The equivalent resistance is:

- (A) 9Ω
- (B) 1Ω
- (C) 3Ω
- (D) 0.33Ω

Q.44

MCQ

Kirchhoff's current law (KCL) is based on conservation of:

- (A) Energy
- (B) Momentum
- (C) Charge
- (D) Mass

Q.45

MCQ

The SI unit of electrical resistivity is:

- (A) Ω
- (B) $\Omega \cdot \text{m}$
- (C) Ω/m
- (D) S/m

Q.46

MCQ

A wire of resistance 10Ω is stretched to double its length. Its new resistance is:

- (A) 5Ω
- (B) 20Ω
- (C) 40Ω
- (D) 80Ω

Q.47

MCQ

The power dissipated in a resistor R carrying current I is:

- (A) IR
- (B) I^2R
- (C) IR^2
- (D) I/R

Q.48

NUMERICAL

What is the terminal voltage of a cell with EMF 2V , internal resistance 0.5Ω , delivering current 2A ?

Ans: _____

Q.49

SHORT

State Kirchhoff's voltage law and apply it to a simple single-loop circuit.

Ans: _____

■ Magnetism

Q.50

MCQ

A charged particle moving parallel to a magnetic field experiences:

- (A) Maximum force
- (B) Zero force
- (C) Centripetal force
- (D) Gravity

Q.51

MCQ

Faraday's law of electromagnetic induction states that induced EMF is proportional to:

- (A) Magnetic field
- (B) Rate of change of magnetic flux
- (C) Current in coil
- (D) Resistance of coil

Q.52

MCQ

Lenz's law is a consequence of conservation of:

- (A) Charge
- (B) Momentum
- (C) Energy
- (D) Mass

Q.53

MCQ

The unit of magnetic flux is:

- (A) Tesla
- (B) Weber
- (C) Ampere
- (D) Henry

Q.54

MCQ

Self-inductance of a coil opposes the change in:

- (A) Voltage
- (B) Resistance
- (C) Current
- (D) Temperature

Q.55

MCQ

A straight conductor carrying current I is placed in a uniform magnetic field B perpendicular to it. The force per unit length on it is:

- (A) B/I
- (B) I/B
- (C) IB
- (D) I^2B

■ Optics

Q.56

MCQ

The focal length of a concave mirror is 15 cm. An object is placed 30 cm in front of it. The image distance is:

- (A) 30 cm
- (B) -30 cm
- (C) 15 cm
- (D) 60 cm

Q.57

MCQ

Total internal reflection occurs when light goes from:

- (A) Rarer to denser medium
- (B) Denser to rarer medium at critical angle
- (C) Any medium at 45°
- (D) Air to glass

Q.58

MCQ

Young's double slit experiment demonstrates:

- (A) Diffraction
- (B) Polarization
- (C) Interference
- (D) Reflection

Q.59

MCQ

Magnifying power of a simple microscope is:

- (A) D/f
- (B) $1 + D/f$
- (C) f/D
- (D) $D/(f+D)$

Q.60

NUMERICAL

A convex lens has focal length 20 cm. An object is placed 30 cm from it. Find the image position.

Ans: _____

Q.61

MCQ

Which colour of light undergoes the maximum deviation when passed through a prism?

- (A) Red
- (B) Green
- (C) Yellow
- (D) Violet

Q.62

NUMERICAL

What is the refractive index of glass if the critical angle is 42° ? ($\sin 42^\circ = 0.669$)

Ans: _____

■ Modern Physics

Q.63

MCQ

According to de Broglie hypothesis, wavelength of a particle is inversely proportional to its:

- (A) Velocity
- (B) Charge
- (C) Momentum
- (D) Energy

Q.64

MCQ

The photoelectric effect was explained by:

- (A) Maxwell
- (B) Hertz
- (C) Einstein
- (D) Planck

Q.65

MCQ

Half-life of a radioactive element is 20 days. After 60 days, the fraction remaining is:

- (A) $1/2$
- (B) $1/4$
- (C) $1/8$
- (D) $1/16$

Q.66

MCQ

In a nuclear fission reaction, energy is released because:

- (A) Neutrons are destroyed
- (B) Mass defect is converted to energy
- (C) Protons are destroyed
- (D) Temperature increases

Q.67

MCQ

The Bohr radius of hydrogen atom ($n=1$) is approximately:

- (A) $0.529 \text{ \AA}$
- (B) $5.29 \text{ \AA}$
- (C) $0.0529 \text{ \AA}$
- (D) $52.9 \text{ \AA}$

Q.68

NUMERICAL

The work function of sodium is 2.3 eV. What is the threshold frequency?

Ans: _____

Q.69

MCQ

Energy of photon with frequency $6 \times 10^{14} \text{ Hz}$ is ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$):

- (A) $3.98 \times 10^{-19} \text{ J}$
- (B) $6 \times 10^{-19} \text{ J}$
- (C) $1 \times 10^{-19} \text{ J}$
- (D) $2 \times 10^{-19} \text{ J}$

Q.70

SHORT

Write the Q-value equation for a nuclear reaction and state what it signifies.

Ans: _____

■ Semiconductors

Q.71

MCQ

The majority carriers in a p-type semiconductor are:

- (A) Electrons
- (B) Holes
- (C) Protons
- (D) Neutrons

Q.72

MCQ

A p-n junction diode in forward bias:

- (A) Has very high resistance
- (B) Allows current to flow easily
- (C) Acts as an insulator
- (D) Blocks current

Q.73

MCQ

A Zener diode is mainly used for:

- (A) Amplification
- (B) Rectification
- (C) Voltage regulation
- (D) Oscillation

Q.74

MCQ

The energy gap in conductors is:

- (A) Large (>3 eV)
- (B) Small (~ 1 eV)
- (C) Zero
- (D) Very large

Q.75

MCQ

In an NPN transistor, the base is:

- (A) Heavily doped
- (B) Lightly doped and thin
- (C) Undoped
- (D) As thick as emitter

■ Current Electricity

Q.76

MCQ

A wire of length 2 m and cross-sectional area 1 mm^2 has resistance 1Ω . Its resistivity is:

- (A) $5 \times 10^{-8} \Omega \cdot \text{m}$
- (B) $2 \times 10^{-8} \Omega \cdot \text{m}$
- (C) $1 \times 10^{-8} \Omega \cdot \text{m}$
- (D) $0.5 \times 10^{-8} \Omega \cdot \text{m}$

■ AC Circuits

Q.77

MCQ

The rms current in an AC circuit with peak current 10 A is:

- (A) 10 A
- (B) 7.07 A
- (C) 5 A
- (D) 14.14 A

Q.78

MCQ

In an LCR series circuit at resonance, the impedance equals:

- (A) 0
- (B) R
- (C) $X_L + X_C$
- (D) $X_L - X_C$

Q.79

MCQ

Power factor of a purely inductive circuit is:

- (A) 1
- (B) 0.5
- (C) 0
- (D) $\sqrt{2}/2$

Q.80

MCQ

The energy stored in an inductor of inductance L carrying current I is:

- (A) LI
- (B) $\frac{1}{2}LI^2$
- (C) LI^2
- (D) L^2I

■ Electromagnetic Waves

Q.81

MCQ

The displacement current was introduced by:

- (A) Faraday
- (B) Maxwell
- (C) Hertz
- (D) Ampere

Q.82

MCQ

Electromagnetic waves are transverse in nature because:

- (A) Electric and magnetic fields are parallel
- (B) E and B are perpendicular to the direction of propagation
- (C) They require a medium
- (D) They travel at speed of sound

Q.83

MCQ

Which electromagnetic wave has the highest frequency?

- (A) Radio waves
- (B) Infrared
- (C) X-rays
- (D) Gamma rays

■ Modern Physics

Q.84

A photon of wavelength 600 nm has energy ($h = 6.63 \times 10^{-34}$ J·s, $c = 3 \times 10^8$ m/s):

Ans: _____

Q.85

MCQ

The number of electrons emitted in photoelectric effect depends on:

- (A) Frequency of light
- (B) Intensity of light
- (C) Wavelength of light
- (D) Work function

Q.86

MCQ

α -particles are identical to the nucleus of:

- (A) Hydrogen
- (B) Helium
- (C) Lithium
- (D) Beryllium

Q.87

MCQ

In β decay, which particle is emitted?

- (A) Proton
- (B) Electron
- (C) Positron
- (D) Neutron

Q.88

MCQ

The mass number of an element changes in which decay?

- (A) β -decay
- (B) γ -decay
- (C) α -decay
- (D) Both B and C

■ Semiconductors

Q.89

MCQ

A logic gate that gives output 1 only when all inputs are 1 is called:

- (A) OR gate
- (B) AND gate
- (C) NOT gate
- (D) NOR gate

Q.90

MCQ

The output of an XOR gate is 0 when:

- (A) Inputs are different
- (B) Inputs are same
- (C) One input is 1
- (D) Both inputs are 0 only

■ Communication Systems

Q.91

MCQ

Amplitude modulation is used in:

- (A) Television broadcasting
- (B) AM radio broadcasting
- (C) Mobile phones
- (D) Radar

Q.92

MCQ

The sky wave propagation is used for communication at frequencies:

- (A) Below 2 MHz
- (B) 2–30 MHz
- (C) Above 40 MHz
- (D) Above 100 MHz

■ Optics

Q.93

SHORT

What is the principle behind optical fiber communication?

Ans: _____

Q.94

SHORT

Explain the formation of rainbow with the help of dispersion of light.

Ans: _____

■ Magnetism

Q.95

NUMERICAL

A particle of charge 2×10^{-16} C moves with velocity 10^6 m/s in a magnetic field of 0.5 T. Find the force on it.

Ans: _____

Q.96

MCQ

Torque acting on a magnetic dipole of moment M in a uniform field B at angle θ is:

- (A) MB
- (B) $MB \sin\theta$
- (C) $MB \cos\theta$
- (D) $MB \tan\theta$

Q.97

MCQ

Diamagnetic materials have relative permeability:

- (A) Greater than 1
- (B) Equal to 1
- (C) Less than 1
- (D) Zero

■ AC Circuits

Q.98

NUMERICAL

In a transformer, if primary turns are 100 and secondary turns are 500, and primary voltage is 220 V, find secondary voltage.

Ans: _____

■ Optics

Q.99

SHORT

The phenomenon of polarisation proves that light is a _____ wave.

Ans: _____

Q.100

MCQ

Resolving power of a microscope increases when the wavelength of light used is:

- (A) Increased
- (B) Decreased
- (C) Unchanged
- (D) Doubled

Q.101

SHORT

State the condition for constructive interference of light.

Ans: _____

■ Waves

Q.102

NUMERICAL

A body of mass 0.5 kg is executing SHM with amplitude 5 cm and frequency 10 Hz. Find the maximum kinetic energy.

Ans: _____

Q.103

MCQ

The speed of sound in air at 27°C is 340 m/s. At 127°C, it would be:

- (A) 392 m/s
- (B) 360 m/s
- (C) 380 m/s
- (D) 420 m/s

Q.104

MCQ

Stationary waves are formed by superposition of two waves travelling in:

- (A) Same direction with same frequency
- (B) Opposite directions with same frequency
- (C) Perpendicular directions
- (D) Any direction

■ Mechanics

Q.105

NUMERICAL

The coefficient of friction between a body and a surface is 0.4. The body weighs 50 N. Find the limiting friction.

Ans: _____

■ Optics

Q.106

MCQ

The ratio of intensities at maxima and minima in a double slit experiment with slit amplitudes a_1 and a_2 is:

- (A) $(a_1 + a_2)^2 / (a_1 - a_2)^2$
- (B) a_1 / a_2
- (C) $(a_1 - a_2) / (a_1 + a_2)$
- (D) a_1^2 / a_2^2

Q.107

MCQ

The angular width of the central maximum in single slit diffraction is:

- (A) λ/d
- (B) $2\lambda/d$
- (C) $\lambda/2d$
- (D) d/λ

■ Mechanics

Q.108

MCQ

The gravitational potential energy of two masses m_1 and m_2 separated by distance r is:

- (A) $-Gm_1m_2/r$
- (B) Gm_1m_2/r
- (C) Gm_1m_2/r^2
- (D) $-Gm_1m_2/r^2$

Q.109

SHORT

A satellite in a circular orbit has kinetic energy equal to _____ times its total energy.

Ans: _____

Q.110

MCQ

Kepler's third law states that $T^2 \propto$:

- (A) r
- (B) r^2
- (C) r^3
- (D) $r^{\frac{3}{2}}$

Q.111

MCQ

When a body is taken from equator to poles, its weight:

- (A) Decreases
- (B) Increases
- (C) Remains same
- (D) Becomes zero

Q.112

MCQ

The bulk modulus of an incompressible liquid is:

- (A) Zero
- (B) Infinity
- (C) 1
- (D) Very small

Q.113

MCQ

Surface tension of a liquid is due to:

- (A) Gravitational force
- (B) Cohesive forces between molecules
- (C) Adhesive forces
- (D) External pressure

Q.114

MCQ

Bernoulli's principle is based on conservation of:

- (A) Mass
- (B) Momentum
- (C) Energy
- (D) Charge

Q.115

MCQ

Thermometric property used in a mercury thermometer is:

- (A) Resistance
- (B) Volume expansion
- (C) EMF
- (D) Pressure

Q.116

MCQ

One calorie equals approximately:

- (A) 1 J
- (B) 4.2 J
- (C) 0.24 J
- (D) 42 J

Q.117

MCQ

The process of heat transfer that does not require a medium is:

- (A) Conduction
- (B) Convection
- (C) Radiation
- (D) Both A and B

Q.118

MCQ

Wien's displacement law relates the peak wavelength of radiation to:

- (A) Frequency
- (B) Temperature
- (C) Intensity
- (D) Energy

Q.119

MCQ

Coefficient of performance of a refrigerator is defined as:

- (A) Work done / Heat removed
- (B) Heat removed / Work done
- (C) Work done / Total heat
- (D) 1 - efficiency

Q.120

MCQ

Degree of freedom for a diatomic gas at room temperature is:

- (A) 3
- (B) 5
- (C) 6
- (D) 7

Q.121

MCQ

The mean free path of a gas molecule increases when:

- (A) Temperature decreases
- (B) Pressure increases
- (C) Temperature increases and pressure decreases
- (D) Density increases

■ Semiconductors

Q.122

MCQ

In a p-type semiconductor, the Fermi level lies:

- (A) At the middle of bandgap
- (B) Near the conduction band
- (C) Near the valence band
- (D) In the conduction band

Q.123

MCQ

The output frequency of a full-wave rectifier for input frequency f is:

- (A) $f/2$
- (B) f
- (C) $2f$
- (D) $4f$

ANSWER KEY — PHYSICS

Q.1: B	Q.2: A	Q.3: A	Q.4: C
Q.5: B	Q.6: A	Q.7: B	Q.8: C
Q.9: C	Q.10: C	Q.11: A	Q.12: B
Q.13: A	Q.14: B	Q.15: A	Q.21: A
Q.22: B	Q.23: B	Q.24: B	Q.26: C
Q.27: B	Q.28: A	Q.29: C	Q.30: C
Q.31: A	Q.34: B	Q.35: D	Q.36: C
Q.37: B	Q.38: B	Q.39: B	Q.40: C
Q.42: D	Q.43: B	Q.44: C	Q.45: B
Q.46: C	Q.47: B	Q.50: B	Q.51: B
Q.52: C	Q.53: B	Q.54: C	Q.55: C
Q.56: A	Q.57: B	Q.58: C	Q.59: B
Q.61: D	Q.63: C	Q.64: C	Q.65: C
Q.66: B	Q.67: A	Q.69: A	Q.71: B
Q.72: B	Q.73: C	Q.74: C	Q.75: B
Q.76: A	Q.77: B	Q.78: B	Q.79: C
Q.80: B	Q.81: B	Q.82: B	Q.83: D
Q.85: B	Q.86: B	Q.87: B	Q.88: C
Q.89: B	Q.90: B	Q.91: B	Q.92: B
Q.96: B	Q.97: C	Q.100: B	Q.103: A
Q.104: B	Q.106: A	Q.107: B	Q.108: A
Q.110: C	Q.111: B	Q.112: B	Q.113: B
Q.114: C	Q.115: B	Q.116: B	Q.117: C
Q.118: B	Q.119: B	Q.120: B	Q.121: C
Q.122: C	Q.123: C		

Note: For SHORT ANSWER and NUMERICAL questions, refer to standard textbooks for detailed solutions.

CHEMISTRY

Atomic Structure

Q.124

MCQ

The number of orbitals in the $n=3$ shell is:

- (A) 3
- (B) 6
- (C) 9
- (D) 12

Q.125

MCQ

Which quantum number determines the shape of an orbital?

- (A) Principal (n)
- (B) Azimuthal (l)
- (C) Magnetic (m)
- (D) Spin (s)

Q.126

MCQ

The electronic configuration of Fe ($Z=26$) is:

- (A) $[\text{Ar}] 3d^6 4s^2$
- (B) $[\text{Ar}] 3d^6$
- (C) $[\text{Ar}] 4s^2 3d^6$
- (D) Both A and C

Q.127

SHORT

Heisenberg's uncertainty principle states that it is impossible to determine simultaneously the exact _____ and _____ of an electron.

Ans: _____

Q.128

MCQ

Which of the following is isoelectronic with N^{2-} ?

- (A) O^{2-}
- (B) F
- (C) Ne
- (D) Na

Q.129

NUMERICAL

The wavelength of an electron with kinetic energy 1.5 eV is ($h = 6.63 \times 10^{-34} \text{ J s}$, $m = 9.1 \times 10^{-31} \text{ kg}$):

Ans: _____

Q.130

MCQ

Aufbau principle states that electrons occupy orbitals in order of:

- (A) Decreasing energy
- (B) Increasing energy
- (C) Spin
- (D) Magnetic quantum number

■ Periodic Table

Q.131

MCQ

Electronegativity across a period:

- (A) Decreases
- (B) Increases
- (C) First increases then decreases
- (D) Remains constant

Q.132

MCQ

Which has the smallest atomic radius?

- (A) Na
- (B) Mg
- (C) Al
- (D) Si

■ Chemical Bonding

Q.133

MCQ

The bond angle in water molecule is:

- (A) 120°
- (B) 109.5°
- (C) 104.5°
- (D) 180°

Q.134

MCQ

Which of the following has a linear shape?

- (A) $\text{H}-\text{O}-\text{H}$
- (B) NH_3
- (C) CO_2
- (D) SO_2

Q.135

MCQ

Hybridization of carbon in ethyne (C_2H_2) is:

- (A) sp^3
- (B) sp^2
- (C) sp
- (D) sp^3d

Q.136

MCQ

Which type of bond is formed by sharing of electrons?

- (A) Ionic
- (B) Covalent
- (C) Metallic
- (D) Hydrogen

Q.137

MCQ

The shape of PCl_5 is:

- (A) Square pyramidal
- (B) Trigonal bipyramidal
- (C) Octahedral
- (D) Tetrahedral

Q.138

SHORT

Explain why BF_3 is a Lewis acid while NF_3 is a Lewis base.

Ans: _____

■ States of Matter

Q.139

MCQ

At STP, 1 mole of ideal gas occupies:

- (A) 22.4 L
- (B) 11.2 L
- (C) 44.8 L
- (D) 2.24 L

Q.140

MCQ

The van der Waals equation accounts for:

- (A) Ideal gas behaviour
- (B) Intermolecular forces and finite molecular volume
- (C) Kinetic energy only
- (D) Pressure only

Q.141

MCQ

Boyle's law states that at constant temperature, pressure is inversely proportional to:

- (A) Temperature
- (B) Moles
- (C) Volume
- (D) Density

■ Thermodynamics (Chem)

Q.142

MCQ

$\Delta G < 0$ for a reaction indicates that the reaction is:

- (A) Non-spontaneous
- (B) At equilibrium
- (C) Spontaneous
- (D) Endothermic

Q.143

MCQ

Hess's law is based on conservation of:

- (A) Mass
- (B) Charge
- (C) Energy
- (D) Momentum

Q.144

MCQ

The standard enthalpy of formation of an element in its standard state is:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Undefined

Q.145

MCQ

Entropy of the universe in a spontaneous process:

- (A) Decreases
- (B) Increases
- (C) Remains same
- (D) Becomes zero

Q.146

NUMERICAL

Calculate ΔG at 300 K if $\Delta H = -30$ kJ/mol and $\Delta S = -100$ J/mol·K.

Ans: _____

■ Equilibrium

Q.147

MCQ

Le Chatelier's principle predicts the shift in equilibrium when:

- (A) A catalyst is added
- (B) Temperature/pressure/concentration is changed
- (C) Volume is constant
- (D) Reaction is at STP

Q.148

MCQ

For a reaction $aA + bB \rightleftharpoons cC + dD$, the equilibrium constant K_c is:

- (A) $[C]^c [D]^d / [A]^a [B]^b$
- (B) $[A][B]/[C][D]$
- (C) $[A]^a [B]^b / [C]^c [D]^d$
- (D) $[C][D]/[A][B]$

Q.149

MCQ

pH of a 0.01 M HCl solution is:

- (A) 1
- (B) 2
- (C) 3
- (D) 7

Q.150

MCQ

Common ion effect suppresses:

- (A) Ionisation of weak acid/base
- (B) Solubility of sparingly soluble salt
- (C) Both A and B
- (D) Reaction rate

Q.151

SHORT

K_p and K_c are related by the equation $K_p = K_c(RT)^{\Delta n}$. What does Δn represent?

Ans: _____

■ Redox

Q.152

MCQ

In the reaction $MnO_2 + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+}$, MnO_2 acts as:

- (A) Reducing agent
- (B) Oxidising agent
- (C) Catalyst
- (D) Solvent

■ Electrochemistry

Q.153

MCQ

Faraday's first law of electrolysis states that mass deposited is proportional to:

- (A) Voltage
- (B) Temperature
- (C) Charge passed
- (D) Resistance

Q.154

MCQ

The standard hydrogen electrode has a potential of:

- (A) +1 V
- (B) -1 V
- (C) 0 V
- (D) +0.5 V

Q.155

NUMERICAL

EMF of a cell is 1.5 V. If Gibbs energy change is $-n \times F \times E_{\text{cell}}$, find ΔG for $n = 2$ ($F = 96500$).

Ans: _____

Q.156

MCQ

In galvanic cells, oxidation occurs at:

- (A) Cathode
- (B) Anode
- (C) Both electrodes
- (D) Salt bridge

■ Chemical Kinetics

Q.157

MCQ

The rate of a first order reaction depends on:

- (A) Concentration²
- (B) Concentration
- (C) Concentration^{1/2}
- (D) Temperature only

Q.158

MCQ

A catalyst increases reaction rate by:

- (A) Increasing ΔH
- (B) Decreasing activation energy
- (C) Increasing temperature
- (D) Changing the products

Q.159

MCQ

Half-life of a first order reaction with rate constant k is:

- (A) $0.693/k$
- (B) $k/0.693$
- (C) $k \times 0.693$
- (D) $1/k$

Q.160

MCQ

The Arrhenius equation relates rate constant to:

- (A) Concentration
- (B) Temperature
- (C) Pressure
- (D) Volume

Q.161

SHORT

Define order and molecularity of a reaction. How do they differ?

Ans: _____

■ Organic Chemistry

Q.162

MCQ

The IUPAC name of $\text{CH}_3\text{-CH(OH)-CH}_3$ is:

- (A) 1-propanol
- (B) 2-propanol
- (C) propan-2-ol
- (D) Both B and C

Q.163

MCQ

Markovnikov's rule applies to addition of HX to:

- (A) Alkanes
- (B) Alkynes only
- (C) Unsymmetrical alkenes
- (D) Aromatic compounds

Q.164

MCQ

The product of dehydration of ethanol is:

- (A) Methanol
- (B) Ethene
- (C) Ethyne
- (D) Diethyl ether (at conc. H_2SO_4 , 170°C)

Q.165

MCQ

Which reagent converts a primary alcohol to an aldehyde?

- (A) PCC
- (B) LiAlH_4
- (C) NaBH_4
- (D) H_2/Ni

Q.166

MCQ

Benzene undergoes preferentially:

- (A) Addition reactions
- (B) Substitution reactions
- (C) Elimination reactions
- (D) Combustion

Q.167

MCQ

Nucleophilic substitution S_N1 is favoured by:

- (A) Primary halide
- (B) Tertiary halide
- (C) Methyl halide
- (D) Vinyl halide

Q.168

MCQ

The Tollens' test is used to identify:

- (A) Ketones
- (B) Alcohols
- (C) Aldehydes
- (D) Carboxylic acids

Q.169

SHORT

Write the reaction mechanism for the nitration of benzene.

Ans: _____

Q.170

MCQ

The ester formed from acetic acid and ethanol in presence of H_2SO_4 is:

- (A) Methyl acetate
- (B) Ethyl acetate
- (C) Propyl acetate
- (D) Ethyl propanoate

Q.171

SHORT

What is a Grignard reagent? Give its general formula and one use.

Ans: _____

Q.172

MCQ

Nylon-6,6 is formed by polymerization of:

- (A) Caprolactam
- (B) Hexamethylenediamine + adipic acid
- (C) Styrene
- (D) Vinyl chloride

Q.173

SHORT

Distinguish between addition polymerization and condensation polymerization with one example each.

Ans: _____

■ p-Block Elements

Q.174

MCQ

The most electronegative element is:

- (A) Oxygen
- (B) Chlorine
- (C) Fluorine
- (D) Nitrogen

Q.175

MCQ

Which noble gas is used in balloons instead of hydrogen?

- (A) Neon
- (B) Argon
- (C) Helium
- (D) Krypton

Q.176

MCQ

Ozone layer is found in:

- (A) Troposphere
- (B) Mesosphere
- (C) Stratosphere
- (D) Thermosphere

■ d-Block Elements

Q.177

MCQ

The highest oxidation state of Mn is:

- (A) +4
- (B) +5
- (C) +7
- (D) +6

Q.178

MCQ

Which transition metal is used as a catalyst in the Haber process?

- (A) Cu
- (B) Fe
- (C) Ni
- (D) Pt

Q.179

SHORT

Why do transition metals form coloured compounds?

Ans: _____

■ Solutions

Q.180

SHORT

Elevation in boiling point is a _____ property that depends on the number of solute particles.

Ans: _____

■ Surface Chemistry

Q.181

MCQ

Which colloid has a dispersed phase as liquid and dispersion medium as solid?

- (A) Foam
- (B) Gel
- (C) Emulsion
- (D) Aerosol

■ Solutions

Q.182

SHORT

Raoult's law states that the vapour pressure of a solution is proportional to the _____ of the solvent.

Ans: _____

■ Surface Chemistry

Q.183

MCQ

Adsorption of gas on solid surface is exothermic due to:

- (A) Entropy increase
- (B) Bond formation at surface
- (C) Expansion of gas
- (D) Cooling effect

■ Solutions

Q.184

SHORT

The vapour pressure of a solution is always _____ that of the pure solvent.

Ans: _____

Q.185

MCQ

Henry's law relates solubility of a gas to:

- (A) Temperature
- (B) Partial pressure of the gas
- (C) Volume of solvent
- (D) Density of solvent

Q.186

Osmotic pressure of a 0.1 M glucose solution at 27°C is ($R = 0.0821 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$):

Ans: _____

Q.187

MCQ

Molarity is defined as moles of solute per:

- (A) 100 g solvent
- (B) 1 kg solvent
- (C) 1 L solution
- (D) 1 L solvent

Q.188

MCQ

Isotonic solutions have the same:

- (A) pH
- (B) Temperature
- (C) Osmotic pressure
- (D) Density

■ Chemistry in Everyday Life

Q.189

MCQ

The degree of hardness of water is due to:

- (A) NaCl
- (B) Ca^{2+} and Mg^{2+} salts
- (C) Iron
- (D) Fluoride

Q.190

SHORT

Soap works by forming a _____ with grease, making it water-soluble.

Ans: _____

Q.191

MCQ

Aspirin is chemically:

- (A) Paracetamol
- (B) Acetylsalicylic acid
- (C) Ibuprofen
- (D) Morphine

Q.192

SHORT

Antacids contain _____ to neutralize excess stomach acid.

Ans: _____

■ Organic Chemistry

Q.193

MCQ

The IUPAC name of acetone is:

- (A) Propanal
- (B) Propanone
- (C) Propanoic acid
- (D) Propanol

Q.194

MCQ

Which test distinguishes between primary, secondary, and tertiary amines?

- (A) Lucas test
- (B) Hinsberg test
- (C) Tollens test
- (D) Baeyer test

Q.195

MCQ

The reaction of carboxylic acid with alcohol in presence of H^+SO_4 is called:

- (A) Saponification
- (B) Esterification
- (C) Hydrolysis
- (D) Condensation

Q.196

MCQ

Gabriel phthalimide synthesis is used to prepare:

- (A) Primary amines
- (B) Secondary amines
- (C) Tertiary amines
- (D) Amides

Q.197

MCQ

Optical isomers exist due to:

- (A) Restricted rotation
- (B) Presence of a chiral carbon
- (C) Double bonds
- (D) Ring structure

Q.198

MCQ

The reaction of alkyl halide with aqueous KOH gives:

- (A) Alkene
- (B) Alcohol
- (C) Ether
- (D) Alkyne

Q.199

SHORT

Dehydration of 2-methyl-2-butanol with H_2SO_4 follows Saytzeff's rule. What is the major product?

Ans: _____

■ Biomolecules

Q.200

MCQ

Amino acids are the monomers of:

- (A) Carbohydrates
- (B) Lipids
- (C) Proteins
- (D) Nucleic acids

Q.201

MCQ

The primary structure of a protein refers to:

- (A) 3D folding
- (B) Sequence of amino acids
- (C) Double helix
- (D) Quaternary structure

Q.202

MCQ

DNA and RNA both contain adenine, guanine, and cytosine. The base unique to DNA is:

- (A) Uracil
- (B) Thymine
- (C) Guanine
- (D) Adenine

Q.203

SHORT

Starch and cellulose are both made of glucose but differ in their _____ linkage.

Ans: _____

■ Coordination Chemistry

Q.204

MCQ

Which coordination compound is used in cancer chemotherapy?

- (A) EDTA
- (B) Cisplatin
- (C) Haemoglobin
- (D) Chlorophyll

Q.205

MCQ

The IUPAC name of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is:

- (A) Tetraaminocopper(II)
- (B) Tetraaminocuprate(II)
- (C) Tetraaminecopper(II) ion
- (D) Copper tetraamine

Q.206

MCQ

Coordination number of Ni in $[\text{Ni}(\text{CO})_4]$ is:

- (A) 2
- (B) 4
- (C) 6
- (D) 8

Q.207

MCQ

The EAN rule (18-electron rule) is most applicable to:

- (A) Ionic compounds
- (B) Organometallic complexes
- (C) Coordination compounds with halide ligands
- (D) Simple salts

Q.208

MCQ

The magnetic moment of Fe^{3+} (d^5 , high spin) is (in BM):

- (A) 1.73
- (B) 3.87
- (C) 4.90
- (D) 5.92

Metallurgy

Q.209

MCQ

Ore used to extract aluminium is:

- (A) Haematite
- (B) Bauxite
- (C) Cassiterite
- (D) Galena

Q.210

MCQ

Zone refining is used to obtain:

- (A) Pure iron
- (B) Ultra-pure semiconductors
- (C) Crude oil fractions
- (D) Aluminium

Q.211

MCQ

Froth flotation process separates ore from gangue based on:

- (A) Density difference
- (B) Magnetic properties
- (C) Preferential wettability
- (D) Chemical reactivity

Q.212

MCQ

The process of converting pig iron to wrought iron is called:

- (A) Smelting
- (B) Calcination
- (C) Puddling
- (D) Roasting

■ Equilibrium

Q.213

MCQ

Which of the following is a strong acid?

- (A) Acetic acid
- (B) Carbonic acid
- (C) Nitric acid
- (D) Phosphoric acid

Q.214

MCQ

Buffer solution resists change in:

- (A) Temperature
- (B) Volume
- (C) pH
- (D) Pressure

Q.215

NUMERICAL

The K_{sp} of AgCl is 1.8×10^{-10} . The molar solubility of AgCl in pure water is:

Ans: _____

Q.216

MCQ

Hydrolysis of a salt of weak acid and strong base gives a solution that is:

- (A) Acidic
- (B) Basic
- (C) Neutral
- (D) Depends on concentration

■ p-Block Elements

Q.217

MCQ

Which halogen is a liquid at room temperature?

- (A) Fluorine
- (B) Chlorine
- (C) Bromine
- (D) Iodine

Q.218

MCQ

Phosphorus exists as P_4 due to:

- (A) High electronegativity
- (B) Tendency to form $p\pi-p\pi$ bonds
- (C) Inability to form $p\pi-p\pi$ bonds readily
- (D) High atomic radius

Q.219

MCQ

The oxide of nitrogen used as laughing gas is:

- (A) NO
- (B) NO_2
- (C) N_2O
- (D) N_2O_5

■ Periodic Table

Q.220

MCQ

The s-block elements are present in groups:

- (A) 1 and 2
- (B) 3 to 12
- (C) 13 to 18
- (D) All of the above

■ s-Block Elements

Q.221

MCQ

Diagonal relationship exists between Li and Mg because of their similar:

- (A) Atomic numbers
- (B) Charge/radius ratio
- (C) Electronic configuration
- (D) Melting points

Q.222

SHORT

What happens when sodium is added to water? Write the balanced equation.

Ans: _____

■ f-Block Elements

Q.223

MCQ

Which of the following lanthanides is used in colour TV screens?

- (A) Ce
- (B) Eu
- (C) La
- (D) Nd

■ Organic Chemistry

Q.224

MCQ

The number of sigma bonds in ethane (C_2H_6) is:

- (A) 6
- (B) 7
- (C) 8
- (D) 5

Q.225

MCQ

Alkanes undergo which type of reaction most readily?

- (A) Addition
- (B) Substitution
- (C) Elimination
- (D) Rearrangement

Q.226

MCQ

Ozonolysis of 2-butene gives:

- (A) 2 molecules of acetaldehyde
- (B) Formaldehyde + propanal
- (C) Acetone + formaldehyde
- (D) 2 ethanol molecules

Q.227

SHORT

The reaction of primary amines with chloroform and KOH is the _____ reaction.

Ans: _____

Q.228

MCQ

Diazotization of primary aromatic amine is done using:

- (A) $\text{NaCl} + \text{HCl}$
- (B) $\text{NaNO}_2 + \text{HCl}$ at $0-5^\circ\text{C}$
- (C) KMnO_4
- (D) H_2SO_4 alone

Q.229

MCQ

Coupling reaction of diazonium salt gives:

- (A) Phenol
- (B) Azo dye
- (C) Amine
- (D) Acid

Q.230

MCQ

Identify the type of isomerism: $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3OCH_3 :

- (A) Chain isomerism
- (B) Position isomerism
- (C) Functional isomerism
- (D) Tautomerism

Q.231

MCQ

The major product of Friedel-Crafts alkylation of benzene with CH_3Cl is:

- (A) Chlorobenzene
- (B) Toluene
- (C) Benzaldehyde
- (D) Benzene

Q.232

MCQ

Kolbe's electrolysis is used to synthesize:

- (A) Acids
- (B) Higher alkanes
- (C) Esters
- (D) Alcohols

■ Electrochemistry

Q.233

MCQ

The concept of electrochemical series helps predict:

- (A) Speed of reaction
- (B) Whether a metal can displace another from its solution
- (C) Solubility
- (D) Colour of compounds

Q.234

MCQ

The standard reduction potential of Zn^{2+}/Zn is -0.76 V and Cu^{2+}/Cu is $+0.34\text{ V}$. The EMF of the Daniell cell is:

- (A) 0.42 V
- (B) 1.10 V
- (C) 0.34 V
- (D) 0.76 V

Q.235

SHORT

Kohlrausch's law of independent migration of ions states that molar conductivity at infinite dilution is the _____ of ionic conductances.

Ans: _____

Q.236

SHORT

The degree of dissociation of a weak electrolyte can be found from the ratio of _____ to _____.

Ans: _____

Q.237

MCQ

The Nernst equation relates electrode potential to:

- (A) Temperature only
- (B) Concentration of ions and standard potential
- (C) Current
- (D) Resistance

■ Materials Chemistry

Q.238

MCQ

Cement is a mixture of calcium silicates and:

- (A) Calcium sulphate
- (B) Calcium carbonate
- (C) Calcium aluminates
- (D) Calcium oxide only

■ Solid State

Q.239

SHORT

Glass is a _____ solid.

Ans: _____

Q.240

MCQ

In an FCC unit cell, the number of atoms per unit cell is:

- (A) 1
- (B) 2
- (C) 4
- (D) 8

Q.241

SHORT

A Schottky defect lowers the _____ of the crystal.

Ans: _____

Q.242

MCQ

The coordination number of Na⁺ in NaCl is:

- (A) 4
 - (B) 6
 - (C) 8
 - (D) 12
-

Q.243

MCQ

The packing efficiency of FCC arrangement is approximately:

- (A) 52%
 - (B) 68%
 - (C) 74%
 - (D) 90%
-

ANSWER KEY — CHEMISTRY

Q.124: C	Q.125: B	Q.126: D	Q.128: C
Q.130: B	Q.131: B	Q.132: D	Q.133: C
Q.134: C	Q.135: C	Q.136: B	Q.137: B
Q.139: A	Q.140: B	Q.141: C	Q.142: C
Q.143: C	Q.144: C	Q.145: B	Q.147: B
Q.148: A	Q.149: B	Q.150: C	Q.152: B
Q.153: C	Q.154: C	Q.156: B	Q.157: B
Q.158: B	Q.159: A	Q.160: B	Q.162: D
Q.163: C	Q.164: B	Q.165: A	Q.166: B
Q.167: B	Q.168: C	Q.170: B	Q.172: B
Q.174: C	Q.175: C	Q.176: C	Q.177: C
Q.178: B	Q.181: B	Q.183: B	Q.185: B
Q.187: C	Q.188: C	Q.189: B	Q.191: B
Q.193: B	Q.194: B	Q.195: B	Q.196: A
Q.197: B	Q.198: B	Q.200: C	Q.201: B
Q.202: B	Q.204: B	Q.205: C	Q.206: B
Q.207: B	Q.208: D	Q.209: B	Q.210: B
Q.211: C	Q.212: C	Q.213: C	Q.214: C
Q.216: B	Q.217: C	Q.218: C	Q.219: C
Q.220: A	Q.221: B	Q.223: B	Q.224: B
Q.225: B	Q.226: A	Q.228: B	Q.229: B
Q.230: C	Q.231: B	Q.232: B	Q.233: B
Q.234: B	Q.237: B	Q.238: C	Q.240: C
Q.242: B	Q.243: C		

Note: For SHORT ANSWER and NUMERICAL questions, refer to standard textbooks for detailed solutions.

■ MATHEMATICS

■ Algebra

Q.244

MCQ

The roots of the quadratic equation $2x^2 - 5x + 3 = 0$ are:

- (A) 1 and $\frac{3}{2}$
- (B) 3 and $\frac{1}{2}$
- (C) $\frac{3}{2}$ and 1
- (D) Both A and C

Q.245

MCQ

If α and β are roots of $x^2 - 5x + 6 = 0$, then $\alpha^2 + \beta^2$ equals:

- (A) 13
- (B) 25
- (C) 11
- (D) 36

Q.246

MCQ

The sum of the first 20 natural numbers is:

- (A) 190
- (B) 200
- (C) 210
- (D) 100

Q.247

MCQ

$\log_{\blacksquare}(8) + \log_{\blacksquare}(4)$ equals:

- (A) 5
- (B) 6
- (C) 7
- (D) 3

Q.248

MCQ

The number of ways to arrange 5 books in a row is:

- (A) 25
- (B) 60
- (C) 120
- (D) 150

■ Probability

Q.249

MCQ

If A and B are two events with $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.1$, then $P(A \cup B)$ is:

- (A) 0.6
- (B) 0.7
- (C) 0.5
- (D) 0.8

■ Algebra

Q.250

MCQ

The 10th term of the AP 3, 7, 11, 15,... is:

- (A) 39
- (B) 43
- (C) 35
- (D) 47

Q.251

MCQ

The sum of infinite GP with first term 3 and common ratio $1/3$ is:

- (A) 4.5
- (B) 3.5
- (C) $9/2$
- (D) 9

Q.252

SHORT

Solve: $|2x - 3| < 7$

Ans: _____

Q.253

NUMERICAL

Find the value of x: $\log_{\blacksquare}(x^2) = \log_{\blacksquare}(100)$

Ans: _____

■ Permutations & Combinations

Q.254

NUMERICAL

How many 4-digit numbers can be formed using digits 1 to 9 without repetition?

Ans: _____

■ Binomial Theorem

Q.255

SHORT

Expand $(x + y)^{\blacksquare}$ using Binomial Theorem.

Ans: _____

Q.256

MCQ

The middle term in the expansion of $(x + 1/x)^{35}$ is:

- (A) 70
- (B) 56
- (C) 28
- (D) 35

■ Trigonometry

Q.257

MCQ

$\sin 30^\circ + \cos 60^\circ$ equals:

- (A) 1
- (B) $\sqrt{2}$
- (C) 0.5
- (D) 1.5

Q.258

MCQ

$\tan(45^\circ) + \cot(45^\circ)$ equals:

- (A) 1
- (B) 2
- (C) $\sqrt{2}$
- (D) 0

Q.259

MCQ

The general solution of $\sin \theta = 1/2$ is:

- (A) $n\pi + \pi/6$
- (B) $n\pi + (-1)^n \pi/6$
- (C) $2n\pi \pm \pi/6$
- (D) $n\pi/6$

Q.260

NUMERICAL

$\cos(A+B) = \cos A \cos B - \sin A \sin B$. Use this to find $\cos 75^\circ$.

Ans: _____

Q.261

MCQ

The principal value of $\arcsin(-1/2)$ is:

- (A) $-\pi/3$
- (B) $-\pi/6$
- (C) $\pi/6$
- (D) $5\pi/6$

Q.262

NUMERICAL

In triangle ABC, if $a = 5$, $b = 7$, $c = 8$, find angle A using cosine rule.

Ans: _____

Q.263

SHORT

Prove that $\sin^2 A + \cos^2 A = 1$.

Ans: _____

Q.264

MCQ

$\tan(2\theta)$ in terms of $\tan \theta$ is:

- (A) $2\tan \theta / (1 - \tan^2 \theta)$
- (B) $2\tan \theta / (1 + \tan^2 \theta)$
- (C) $\tan^2 \theta / 2$
- (D) $1 / (2\tan \theta)$

■ Coordinate Geometry

Q.265

MCQ

The distance between points (3, 4) and (0, 0) is:

- (A) 5
- (B) 7
- (C) 4
- (D) 3

Q.266

MCQ

The slope of the line joining (1, 2) and (3, 8) is:

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Q.267

MCQ

Equation of a circle with centre (2, -3) and radius 5 is:

- (A) $(x-2)^2 + (y+3)^2 = 5$
- (B) $(x-2)^2 + (y+3)^2 = 25$
- (C) $(x+2)^2 + (y-3)^2 = 25$
- (D) $x^2 + y^2 = 25$

Q.268

MCQ

The eccentricity of a circle is:

- (A) 0
- (B) 1
- (C) Between 0 and 1
- (D) > 1

Q.269

MCQ

The foci of the ellipse $x^2/25 + y^2/9 = 1$ are at:

- (A) $(\pm 4, 0)$
- (B) $(\pm 3, 0)$
- (C) $(\pm 5, 0)$
- (D) $(0, \pm 4)$

Q.270

SHORT

Find the equation of line passing through (2, 3) with slope -1.

Ans: _____

Q.271

MCQ

The angle between lines $y = x$ and $y = \sqrt{3}x$ is:

- (A) 15°
- (B) 30°
- (C) 45°
- (D) 60°

Q.272

MCQ

The vertex of the parabola $y^2 = 8x$ is at:

- (A) (0, 0)
- (B) (2, 0)
- (C) (0, 2)
- (D) (4, 0)

■ Calculus

Q.273

MCQ

$\lim_{x \rightarrow 0} (\sin x)/x$ equals:

- (A) 0
- (B) ∞
- (C) 1
- (D) Undefined

Q.274

MCQ

The derivative of $x^3 + 3x^2 - x + 5$ is:

- (A) $3x^2 + 6x + 1$
- (B) $3x^2 + 6x - 1$
- (C) $x^2 + 6x - 1$
- (D) $3x + 6x - 1$

Q.275

MCQ

$\int x^2 dx$ equals:

- (A) $2x$
- (B) $x^3/3 + C$
- (C) $x^3 + C$
- (D) $3x^3 + C$

Q.276

MCQ

The derivative of $\sin(x^2)$ is:

- (A) $\cos(x^2)$
- (B) $2x \cos(x^2)$
- (C) $\cos(2x)$
- (D) $2\cos(x^2)$

Q.277

MCQ

If $f(x) = e^{\sin x}$, then $f'(x)$ equals:

- (A) $e^{\sin x} \cos x$
- (B) $e^{\sin x}$
- (C) $x e^{\sin x} \cos x$
- (D) $e^{\sin x}/x$

Q.278

MCQ

The value of $\int_0^1 x dx$ is:

- (A) 0
- (B) $1/2$
- (C) 1
- (D) 2

Q.279

MCQ

A function $f(x)$ has a local maximum where $f'(x) = 0$ and $f''(x)$ is:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Undefined

Q.280

NUMERICAL

Find dy/dx if $y = \ln(3x^2 + 1)$.

Ans: _____

Q.281

NUMERICAL

Evaluate $\int \sin x \cos x dx$.

Ans: _____

Q.282

MCQ

The area enclosed between $y = x^2$ and $y = x$ is:

- (A) $1/6$
- (B) $1/3$
- (C) $1/2$
- (D) 1

Q.283

NUMERICAL

If $y = x \cdot e^x$, find d^2y/dx^2 .

Ans: _____

■ Vectors

Q.284

MCQ

The magnitude of vector $3i + 4j$ is:

- (A) 5
- (B) 7
- (C) 1
- (D) 25

Q.285

MCQ

The dot product of i and j is:

- (A) 1
- (B) 0
- (C) i
- (D) -1

Q.286

MCQ

If vectors a and b are perpendicular, then $a \cdot b$ equals:

- (A) 1
- (B) $|a||b|$
- (C) 0
- (D) -1

Q.287

MCQ

The cross product of a vector with itself is:

- (A) 1
- (B) The vector itself
- (C) Zero vector
- (D) Unit vector

Q.288

NUMERICAL

Find the unit vector in the direction of $2i - j + 2k$.

Ans: _____

■ 3D Geometry

Q.289

MCQ

The direction cosines of the x-axis are:

- (A) (1, 0, 0)
- (B) (0, 1, 0)
- (C) (0, 0, 1)
- (D) (1, 1, 0)

Q.290

MCQ

The equation of a plane parallel to the XY-plane passing through (0, 0, 5) is:

- (A) $x = 5$
- (B) $y = 5$
- (C) $z = 5$
- (D) $x + y + z = 5$

■ Matrices

Q.291

MCQ

The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is:

- (A) $ab - cd$
- (B) $ad + bc$
- (C) $ad - bc$
- (D) $ac - bd$

Q.292

MCQ

A square matrix A is invertible if and only if:

- (A) A is symmetric
- (B) $\det(A) \neq 0$
- (C) A is diagonal
- (D) A is upper triangular

Q.293

MCQ

The transpose of matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is:

- (A) $\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
- (C) $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$
- (D) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Q.294

NUMERI
CAL

Solve the system of equations using matrices: $2x + y = 5$, $x + 3y = 10$.

Ans: _____

■ Probability

Q.295

MCQ

Two fair dice are thrown. The probability of getting a sum of 7 is:

- (A) $\frac{1}{6}$
- (B) $\frac{1}{7}$
- (C) $\frac{5}{36}$
- (D) $\frac{7}{36}$

■ Statistics

Q.296

MCQ

Mean of the data 4, 7, 3, 8, 5, 9, 2 is:

- (A) 5
- (B) 5.4
- (C) 5.5
- (D) 6

Q.297

MCQ

Variance is the square of:

- (A) Mean
- (B) Standard deviation
- (C) Range
- (D) Median

■ Probability

Q.298

MCQ

Bayes' theorem is used to find:

- (A) Joint probability
- (B) Conditional probability
- (C) Marginal probability
- (D) Total probability

Q.299

NUMERICAL

A bag has 3 red and 4 blue balls. Two are drawn without replacement. Find the probability both are red.

Ans: _____

■ Functions

Q.300

SHORT

If $f(x) = x^2 - 4x + 3$, find the values of x where $f(x) < 0$.

Ans: _____

Q.301

MCQ

The domain of $f(x) = \sqrt{4 - x^2}$ is:

- (A) $[-2, 2]$
- (B) $(-\infty, -2) \cup (2, \infty)$
- (C) $(-2, 2)$
- (D) $[-4, 4]$

Q.302

MCQ

A function f is one-to-one (injective) if:

- (A) $f(a) = f(b) \Rightarrow a = b$
- (B) Every y has a preimage
- (C) f is continuous
- (D) f is differentiable

Q.303

MCQ

The inverse of $f(x) = 2x + 3$ is:

- (A) $(x-3)/2$
- (B) $(x+3)/2$
- (C) $2x-3$
- (D) $1/(2x+3)$

■ Complex Numbers

Q.304

NUMERICAL

Find the complex number $(3 + 4i)(3 - 4i)$.

Ans: _____

Q.305

MCQ

The modulus of $5 + 12i$ is:

- (A) 13
- (B) 17
- (C) 5
- (D) 12

Q.306

MCQ

Argument of the complex number $-1 + i$ is:

- (A) $\pi/4$
- (B) $3\pi/4$
- (C) $-\pi/4$
- (D) $\pi/2$

Q.307

MCQ

The cube roots of unity sum to:

- (A) 1
- (B) -1
- (C) 0
- (D) 3

Q.308

SHORT

If $|z| = 1$, z lies on a _____ in the Argand plane.

Ans: _____

Q.309

NUMERICAL

Solve: $x^2 - 6x + 13 = 0$ (roots are complex)

Ans: _____

■ Binomial Theorem

Q.310

MCQ

The number of terms in the expansion of $(a + b)^n$ is:

- (A) n
- (B) $n-1$
- (C) $n+1$
- (D) $2n$

Q.311

MCQ

The coefficient of x^3 in the expansion of $(1 + x)^{15}$ is:

- (A) 5
- (B) 10
- (C) 15
- (D) 20

Q.312

NUMERICAL

The 5th term in the expansion of $(2x - 3)^{15}$ is:

Ans: _____

■ Permutations & Combinations

Q.313

NUMERICAL

In how many ways can 4 boys and 3 girls be arranged in a row so that no two girls are adjacent?

Ans: _____

Q.314

MCQ

The number of handshakes in a group of 10 people (each shakes hands with everyone else once) is:

- (A) 45
- (B) 90
- (C) 100
- (D) 50

■ Sequences & Series

Q.315

NUMERICAL

The sum of an infinite GP is 8 and the first term is 2. Find the common ratio.

Ans: _____

Q.316

MCQ

The 4th term of a GP with first term 3 and common ratio 2 is:

- (A) 12
- (B) 24
- (C) 48
- (D) 6

Q.317

MCQ

Sum of first n terms of AP is $\frac{n}{2} \times (\text{first} + \text{last})$. Sum of first 10 terms of AP 2, 5, 8,... is:

- (A) 155
- (B) 145
- (C) 135
- (D) 165

Q.318

NUMERICAL

Find the sum: $1 + 2 + 3 + \dots + 100$.

Ans: _____

■ Calculus

Q.319

MCQ

The derivative of $\tan x$ is:

- (A) $\sec x$
- (B) $\sec^2 x$
- (C) $\operatorname{cosec}^2 x$
- (D) $-\sec^2 x$

Q.320

MCQ

$\int e^{2x} dx$ equals:

- (A) $2e^{2x} + C$
- (B) $e^{2x}/2 + C$
- (C) $e^{2x} + C$
- (D) $2xe^{2x} + C$

Q.321

MCQ

The maximum value of $f(x) = -x^2 + 4x + 1$ is:

- (A) 5
- (B) 4
- (C) 3
- (D) 6

Q.322

MCQ

$\int_0^\pi \sin x dx$ equals:

- (A) 0
- (B) 2
- (C) π
- (D) 1

Q.323

MCQ

The area under the curve $y = x^2$ from $x = 0$ to $x = 3$ is:

- (A) 9
- (B) 18
- (C) 27
- (D) 3

Q.324

MCQ

dy/dx for $y = x^n$ is:

- (A) $nx^{(n-1)}$
- (B) $x^{(n+1)}/(n+1)$
- (C) $n/x^{(n-1)}$
- (D) $x^n \cdot n$

Q.325

SHORT

The equation of the tangent to $y = x^2$ at $x = 2$ is:

Ans: _____

Q.326

SHORT

Verify Rolle's theorem for $f(x) = x^2 - 4$ on $[-2, 2]$.

Ans: _____

Q.327

NUMERI
CAL

If $y = \sin x$, find d^2y/dx^2 .

Ans: _____

Q.328

MCQ

Evaluate: $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$

- (A) 4
- (B) 0
- (C) ∞
- (D) 2

Q.329

MCQ

The value of $\int_{-1}^1 (x^2 + 1) dx$ is:

- (A) $4/3$
- (B) 2
- (C) $3/2$
- (D) $5/3$

■ Coordinate Geometry

Q.330

MCQ

Two lines are parallel if their slopes are:

- (A) Negative reciprocals
- (B) Equal
- (C) Both zero
- (D) Undefined

Q.331

NUMERI
CAL

The perpendicular distance of point (3, 4) from the line $3x - 4y + 5 = 0$ is:

Ans: _____

Q.332

MCQ

The locus of a point equidistant from two fixed points is the:

- (A) Circle
- (B) Parabola
- (C) Perpendicular bisector
- (D) Ellipse

Q.333

MCQ

The centre of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ is:

- (A) (3, -2)
- (B) (-3, 2)
- (C) (6, -4)
- (D) (-6, 4)

Q.334

NUMERI
CAL

The sum of the squares of the intercepts of the line $3x + 4y = 12$ on the axes is:

Ans: _____

■ Probability

Q.335

NUMERI
CAL

If $P(A) = 0.3$, $P(B) = 0.4$, and A and B are independent, find $P(A \cap B)$.

Ans: _____

Q.336

MCQ

In binomial distribution $B(n, p)$, the mean is:

- (A) np
- (B) npq
- (C) $\sqrt{npq}$
- (D) p/n

Q.337

SHORT

Poisson distribution is a limiting form of binomial distribution when n is _____ and p is _____.

Ans: _____

Q.338

NUMERI
CAL

A coin is tossed 3 times. Find the probability of getting exactly 2 heads.

Ans: _____

■ Statistics

Q.339

NUMERI
CAL

The standard deviation of data 2, 4, 6, 8, 10 is:

Ans: _____

■ Matrices

Q.340

MCQ

If a matrix A has order 3×3 and $\det(A) = 2$, then $\det(3A)$ is:

- (A) 6
- (B) 18
- (C) 54
- (D) 2

Q.341

MCQ

The rank of the zero matrix is:

- (A) 0
- (B) 1
- (C) n
- (D) Undefined

Q.342

MCQ

The value of $\det[[1,2,3],[4,5,6],[7,8,9]]$ is:

- (A) 0
- (B) 1
- (C) -1
- (D) 3

■ Vectors

Q.343

MCQ

The projection of vector a on vector b is:

- (A) $a \times b$
- (B) $a \cdot b / |b|$
- (C) $|a||b| \sin\theta$
- (D) $a \cdot b / |a|$

Q.344

NUMERICAL

If $a = 2i + 3j - k$ and $b = i - j + 2k$, find $a \cdot b$.

Ans: _____

Q.345

MCQ

The area of parallelogram formed by vectors a and b is:

- (A) $a \cdot b$
- (B) $|a \times b|$
- (C) $|a| + |b|$
- (D) $a \cdot b / 2$

■ 3D Geometry

Q.346

MCQ

The angle between lines with direction ratios (1, 1, 0) and (0, 1, 1) is:

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Q.347

NUMERICAL

The distance between parallel planes $2x + y - 2z = 5$ and $2x + y - 2z = 8$ is:

Ans: _____

Q.348

MCQ

The equation of a sphere with centre (1, 2, 3) and radius 4 is:

- (A) $(x-1)^2 + (y-2)^2 + (z-3)^2 = 16$
- (B) $(x-1)^2 + (y-2)^2 + (z-3)^2 = 4$
- (C) $(x+1)^2 + (y+2)^2 + (z+3)^2 = 16$
- (D) $x^2 + y^2 + z^2 = 16$

■ Inverse Trigonometry

Q.349

MCQ

$\sin^{-1}(x) + \cos^{-1}(x)$ equals:

- (A) π
- (B) $\pi/4$
- (C) $\pi/2$
- (D) 0

Q.350

MCQ

The value of $\tan^{-1}(1) + \tan^{-1}(\sqrt{3})$ is:

- (A) $\pi/4$
- (B) $\pi/3$
- (C) $7\pi/12$
- (D) $5\pi/12$

Q.351

MCQ

The principal range of $\sin^{-1}(x)$ is:

- (A) $[0, \pi]$
- (B) $[-\pi/2, \pi/2]$
- (C) $(-\pi, \pi)$
- (D) $[0, 2\pi]$

■ Trigonometry

Q.352

MCQ

The number of solutions of $\sin x = x/3$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) Infinite

Q.353

NUMERI
CAL

In a triangle, if $a = 5$, $b = 6$, and $C = 60^\circ$, find side c using cosine rule.

Ans: _____

Q.354

MCQ

The maximum value of $\sin x + \cos x$ is:

- (A) 1
- (B) $\sqrt{2}$
- (C) 2
- (D) $\sqrt{3}$

Q.355

MCQ

$\sin(A + B) - \sin(A - B)$ simplifies to:

- (A) $2\sin A \cos B$
- (B) $2\cos A \sin B$
- (C) $2\sin B \cos A$
- (D) Both B and C

■ Algebra

Q.356

MCQ

If $\log_a b = x$, then a^x equals:

- (A) x
- (B) b
- (C) a
- (D) 1

Q.357

NUMERI
CAL

Solve: $2^{(x+1)} = 32$

Ans: _____

Q.358

MCQ

The number of real roots of the equation $x^4 - 5x^2 + 4 = 0$ is:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q.359

MCQ

If α, β, γ are roots of $x^3 - 6x^2 + 11x - 6 = 0$, find $\alpha + \beta + \gamma$.

- (A) 6
- (B) -11
- (C) -6
- (D) 11

■ Functions

Q.360

SHORT

The graph of $y = |x|$ has a _____ at the origin.

Ans: _____

■ Sets & Relations

Q.361

MCQ

The number of elements in the power set of $\{1, 2, 3, 4\}$ is:

- (A) 8
- (B) 12
- (C) 16
- (D) 24

Q.362

MCQ

If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \cap B$ is:

- (A) $\{1, 2, 3, 4, 5\}$
- (B) $\{3\}$
- (C) $\{\}$
- (D) $\{1, 2\}$

Q.363

MCQ

A relation R on set A is called equivalence relation if it is:

- (A) Reflexive only
- (B) Symmetric only
- (C) Reflexive, symmetric, and transitive
- (D) Transitive only

■ Calculus

Q.364

MCQ

$\int (1/x) dx$ equals:

- (A) $1/x^2 + C$
- (B) $\ln|x| + C$
- (C) $-1/x + C$
- (D) $x + C$

■ Differential Equations

Q.365

MCQ

The differential equation $dy/dx = y/x$ can be solved by:

- (A) Linear method
- (B) Variable separable method
- (C) Bernoulli method
- (D) Exact method

Q.366

MCQ

The order of the differential equation $d^2y/dx^2 + (dy/dx)^3 = y$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q.367

NUMERICAL

General solution of $dy/dx = e^x$ is:

Ans: _____

Q.368

MCQ

The degree of the differential equation $(d^2y/dx^2)^2 + (dy/dx)^3 = 5x$ is:

- (A) 2
- (B) 3
- (C) 1
- (D) 6

■ Coordinate Geometry

Q.369

MCQ

A line perpendicular to $ax + by + c = 0$ has slope:

- (A) $-a/b$
- (B) b/a
- (C) a/b
- (D) $-b/a$

Q.370

MCQ

The reflection of point (3, -5) in the x-axis is:

- (A) (-3, 5)
- (B) (3, 5)
- (C) (-3, -5)
- (D) (5, 3)

Q.371

MCQ

The midpoint of (4, 6) and (2, -2) is:

- (A) (3, 2)
- (B) (3, 4)
- (C) (6, 4)
- (D) (2, 3)

■ Algebra

Q.372

MCQ

The number of integers between -3 and 3 (inclusive) satisfying $|x| \leq 2$ is:

- (A) 3
- (B) 5
- (C) 4
- (D) 7

■ Mathematical Induction

Q.373

SHORT

Prove by induction: $1 + 2 + 3 + \dots + n = n(n+1)/2$

Ans: _____

■ Sets & Relations

Q.374

SHORT

For sets A and B, $n(A \cup B) = n(A) + n(B) - \underline{\hspace{2cm}}$.

Ans: _____

■ Probability

Q.375

NUMERICAL

If $P(A) = 0.5$ and $P(B|A) = 0.4$, find $P(A \cap B)$.

Ans: _____

■ Statistics

Q.376

SHORT

Normal distribution is also called the _____ curve.

Ans: _____

ANSWER KEY — MATHEMATICS

Q.244: D	Q.245: A	Q.246: C	Q.247: A
Q.248: C	Q.249: A	Q.250: A	Q.251: A
Q.256: A	Q.257: A	Q.258: B	Q.259: B
Q.261: B	Q.264: A	Q.265: A	Q.266: B
Q.267: B	Q.268: A	Q.269: A	Q.271: A
Q.272: A	Q.273: C	Q.274: B	Q.275: B
Q.276: B	Q.277: B	Q.278: B	Q.279: B
Q.282: A	Q.284: A	Q.285: B	Q.286: C
Q.287: C	Q.289: A	Q.290: C	Q.291: C
Q.292: B	Q.293: B	Q.295: A	Q.296: B
Q.297: B	Q.298: B	Q.301: A	Q.302: A
Q.303: A	Q.305: A	Q.306: B	Q.307: C
Q.310: C	Q.311: B	Q.314: A	Q.316: B
Q.317: B	Q.319: B	Q.320: B	Q.321: A
Q.322: B	Q.323: A	Q.324: A	Q.328: A
Q.329: A	Q.330: B	Q.332: C	Q.333: A
Q.336: A	Q.340: C	Q.341: A	Q.342: A
Q.343: B	Q.345: B	Q.346: C	Q.348: A
Q.349: C	Q.350: C	Q.351: B	Q.352: C
Q.354: B	Q.355: D	Q.356: B	Q.358: C
Q.359: A	Q.361: C	Q.362: B	Q.363: C
Q.364: B	Q.365: B	Q.366: B	Q.368: A
Q.369: B	Q.370: B	Q.371: A	Q.372: B

Note: For SHORT ANSWER and NUMERICAL questions, refer to standard textbooks for detailed solutions.

■ BIOLOGY

■ Cell Biology

Q.377

MCQ

The powerhouse of the cell is:

- (A) Nucleus
- (B) Ribosome
- (C) Mitochondria
- (D) Golgi apparatus

Q.378

MCQ

DNA replication is:

- (A) Conservative
- (B) Semi-conservative
- (C) Dispersive
- (D) None

Q.379

MCQ

Ribosomes are the site of:

- (A) DNA synthesis
- (B) Protein synthesis
- (C) Lipid synthesis
- (D) ATP synthesis

Q.380

MCQ

Cell wall in plants is composed of:

- (A) Chitin
- (B) Cellulose
- (C) Peptidoglycan
- (D) Murein

Q.381

MCQ

Which organelle is absent in animal cells?

- (A) Mitochondria
- (B) Ribosome
- (C) Chloroplast
- (D) Endoplasmic reticulum

Q.382

MCQ

The fluid mosaic model of cell membrane was proposed by:

- (A) Watson and Crick
- (B) Singer and Nicolson
- (C) Darwin and Wallace
- (D) Schleiden and Schwann

Q.383

MCQ

During mitosis, chromosomes align at the metaphase plate in:

- (A) Prophase
- (B) Metaphase
- (C) Anaphase
- (D) Telophase

Q.384

SHORT

Distinguish between prokaryotic and eukaryotic cells. Give two examples of each.

Ans: _____

Q.385

SHORT

What is the significance of meiosis in sexually reproducing organisms?

Ans: _____

■ Genetics

Q.386

MCQ

Mendel's law of segregation states that alleles separate during:

- (A) Fertilization
- (B) Gamete formation
- (C) Mitosis
- (D) Interphase

Q.387

MCQ

A cross between Tt and Tt will produce offspring in ratio:

- (A) 3:1
- (B) 1:2:1
- (C) 1:1
- (D) 2:1

Q.388

MCQ

In incomplete dominance, F₁ of Red × White cross is:

- (A) Red
- (B) White
- (C) Pink
- (D) Spotted

Q.389

MCQ

Colour blindness is a sex-linked recessive disorder carried on the:

- (A) Autosome
- (B) Y chromosome
- (C) X chromosome
- (D) Mitochondrial DNA

Q.390

NUMERICAL

A person with genotype $X^{h}X$ (carrier) for haemophilia mates with a normal male (XY). What proportion of sons will be haemophilic?

Ans: _____

Q.391

MCQ

The central dogma of molecular biology is:

- (A) DNA $\rightarrow$ RNA $\rightarrow$ Protein
- (B) RNA $\rightarrow$ DNA $\rightarrow$ Protein
- (C) Protein $\rightarrow$ RNA $\rightarrow$ DNA
- (D) DNA $\rightarrow$ Protein $\rightarrow$ RNA

Q.392

MCQ

Point mutation involving replacement of one base by another is called:

- (A) Frameshift mutation
- (B) Substitution mutation
- (C) Deletion
- (D) Insertion

Q.393

SHORT

Explain co-dominance with the example of ABO blood groups.

Ans: _____

■ Evolution

Q.394

MCQ

Darwin's theory of natural selection states that organisms with better adaptations have:

- (A) Higher mutation rate
- (B) Greater reproductive success
- (C) Larger body size
- (D) More varied diet

Q.395

MCQ

Analogous organs are those that have:

- (A) Same origin, different function
- (B) Different origin, same function
- (C) Same origin and function
- (D) Different origin and function

Q.396

MCQ

The founder effect is a type of:

- (A) Natural selection
- (B) Genetic drift
- (C) Gene flow
- (D) Mutation

Q.397

MCQ

The Hardy-Weinberg principle states that allele frequencies remain constant in the absence of:

- (A) Sexual reproduction
- (B) Evolutionary forces
- (C) Meiosis
- (D) Fertilization

Q.398

MCQ

Miller-Urey experiment demonstrated that:

- (A) Life came from space
- (B) Amino acids could form from inorganic compounds
- (C) DNA is the genetic material
- (D) Cells arise spontaneously

■ Human Physiology

Q.399

MCQ

The hormone responsible for regulating blood sugar is:

- (A) Glucagon
- (B) Insulin
- (C) Thyroxine
- (D) Cortisol

Q.400

MCQ

Which blood type is the universal donor?

- (A) A
- (B) B
- (C) AB
- (D) O

Q.401

MCQ

The functional unit of the kidney is:

- (A) Nephron
- (B) Glomerulus
- (C) Loop of Henle
- (D) Bowman's capsule

Q.402

MCQ

The sinoatrial node is responsible for:

- (A) Blood cell production
- (B) Heart beat initiation
- (C) Hormone secretion
- (D) Clotting

Q.403

MCQ

The partial pressure of O_2 in alveoli is approximately:

- (A) 40 mmHg
- (B) 104 mmHg
- (C) 760 mmHg
- (D) 160 mmHg

Q.404

SHORT

Name the enzymes involved in DNA replication and describe the role of each.

Ans: _____

Q.405

SHORT

Which hormone is called the 'fight or flight' hormone and why?

Ans: _____

Q.406

SHORT

Describe the mechanism of nerve impulse transmission across a synapse.

Ans: _____

Q.407

MCQ

The RBC count in a healthy adult male is approximately:

- (A) 2-3 million/ mm^3
- (B) 4.5-5.5 million/ mm^3
- (C) 7-8 million/ mm^3
- (D) 1-2 million/ mm^3

Q.408

SHORT

Bile juice is secreted by the _____ and stored in the _____.

Ans: _____

■ Plant Physiology

Q.409

MCQ

C4 plants have an advantage over C3 plants because they:

- (A) Produce more glucose per CO_2 fixed
- (B) Minimize photorespiration
- (C) Grow in cold climates
- (D) Require less light

Q.410

MCQ

The process by which plants lose water through their leaves is called:

- (A) Osmosis
- (B) Transpiration
- (C) Guttation
- (D) Translocation

Q.411

MCQ

Gibberellins promote:

- (A) Root growth
- (B) Stem elongation and seed germination
- (C) Leaf senescence
- (D) Fruit ripening

Q.412

MCQ

The Calvin cycle fixes CO_2 in the:

- (A) Thylakoid membrane
- (B) Stroma
- (C) Cytoplasm
- (D) Mitochondria

Q.413

SHORT

Describe the light reactions of photosynthesis, including where they occur and what products are formed.

Ans: _____

■ Ecology

Q.414

MCQ

The 10% law of energy transfer in a food chain was given by:

- (A) Darwin
- (B) Lindemann
- (C) Odum
- (D) Haeckel

Q.415

MCQ

A biome is characterised by its:

- (A) Soil type only
- (B) Dominant vegetation and climate
- (C) Animal species only
- (D) Water availability

Q.416

MCQ

Which of the following is a primary consumer?

- (A) Lion
- (B) Frog
- (C) Grasshopper
- (D) Snake

Q.417

MCQ

Biodiversity hotspots are regions with:

- (A) High species diversity only
- (B) High endemism and significant habitat loss
- (C) Large forest cover
- (D) Only endangered species

Q.418

SHORT

The greenhouse gases include CO₂, methane, and _____ in the atmosphere.

Ans: _____

Q.419

MCQ

Mutualism, commensalism, and parasitism are all types of:

- (A) Competition
- (B) Symbiosis
- (C) Predation
- (D) Amensalism

Q.420

SHORT

Explain why deforestation leads to decreased rainfall in nearby regions.

Ans: _____

■ Reproduction

Q.421

MCQ

The male gamete in flowering plants is formed in:

- (A) Ovule
- (B) Anther
- (C) Style
- (D) Petal

Q.422

MCQ

Double fertilization is unique to:

- (A) Gymnosperms
- (B) Angiosperms
- (C) Pteridophytes
- (D) Bryophytes

Q.423

MCQ

In humans, the developing embryo implants in the:

- (A) Ovary
- (B) Fallopian tube
- (C) Uterine wall (endometrium)
- (D) Cervix

Q.424

SHORT

What is parthenogenesis? Give an example in animals.

Ans: _____

■ Biotechnology

Q.425

MCQ

Restriction enzymes cut DNA at specific sequences called:

- (A) Reading frames
- (B) Palindromic sequences
- (C) Promoters
- (D) Exons

Q.426

MCQ

PCR (Polymerase Chain Reaction) is used to:

- (A) Sequence proteins
- (B) Amplify specific DNA sequences
- (C) Translate mRNA
- (D) Cleave DNA

Q.427

MCQ

The first transgenic crop approved for use was:

- (A) Bt cotton
- (B) Golden rice
- (C) Flavr Savr tomato
- (D) Roundup Ready soy

Q.428

SHORT

Explain the role of plasmids as vectors in recombinant DNA technology.

Ans: _____

Q.429

MCQ

ELISA (Enzyme-Linked Immunosorbent Assay) is based on:

- (A) Gel electrophoresis
- (B) Antigen-antibody interaction
- (C) PCR amplification
- (D) DNA sequencing

■ Diversity of Life

Q.430

MCQ

Which kingdom includes organisms with cell walls made of chitin?

- (A) Monera
- (B) Plantae
- (C) Fungi
- (D) Protista

Q.431

MCQ

The binomial nomenclature system was introduced by:

- (A) Darwin
- (B) Mendel
- (C) Linnaeus
- (D) Lamarck

Q.432

MCQ

Viruses are not considered living organisms primarily because they:

- (A) Have no DNA
- (B) Cannot reproduce without a host cell
- (C) Lack a cell membrane
- (D) Have no proteins

Q.433

SHORT

NCERT defines five kingdoms. Name them and give one distinguishing feature of each.

Ans: _____

■ Cell Biology

Q.434

MCQ

The instrument used to study living cells is called a:

- (A) Spectrophotometer
- (B) Electron microscope
- (C) Light microscope
- (D) Chromatograph

Q.435

SHORT

ATP (adenosine triphosphate) is the currency of _____ in all living cells.

Ans: _____

Q.436

MCQ

Glycolysis occurs in the:

- (A) Mitochondria
- (B) Nucleus
- (C) Cytoplasm
- (D) Ribosome

Q.437

MCQ

Net ATP produced during glycolysis (per glucose) is:

- (A) 38
- (B) 36
- (C) 2
- (D) 4

Q.438

SHORT

The Krebs cycle occurs in the _____ of mitochondria.

Ans: _____

Q.439

MCQ

Electron transport chain is located on the:

- (A) Outer mitochondrial membrane
- (B) Inner mitochondrial membrane
- (C) Matrix
- (D) Cytoplasm

■ Genetics

Q.440

MCQ

The enzyme that unwinds the DNA double helix during replication is:

- (A) DNA polymerase
- (B) Ligase
- (C) Helicase
- (D) Primase

Q.441

SHORT

mRNA is read in the _____ direction during translation.

Ans: _____

Q.442

MCQ

The codon AUG codes for:

- (A) Stop
- (B) Methionine (start)
- (C) Valine
- (D) Lysine

Q.443

MCQ

How many codons code for stop signals?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

■ Biotechnology

Q.444

MCQ

Gene cloning requires all of the following EXCEPT:

- (A) Restriction enzyme
- (B) Vector
- (C) Ligase
- (D) Ribozyme

Q.445

MCQ

The Ti plasmid is used as a vector to introduce genes into:

- (A) Bacteria
- (B) Yeast
- (C) Plants
- (D) Animals

Q.446

MCQ

The first hormone produced by recombinant DNA technology for commercial use was:

- (A) Thyroxine
- (B) Insulin
- (C) Cortisol
- (D) Glucagon

Q.447

MCQ

Gel electrophoresis separates DNA fragments based on:

- (A) Charge only
- (B) Size and charge
- (C) Sequence
- (D) Base composition

Q.448

MCQ

CRISPR-Cas9 is used for:

- (A) PCR amplification
- (B) Gene editing
- (C) Protein synthesis
- (D) Electrophoresis

Q.449

MCQ

Stem cells are unique because they can:

- (A) Only divide
- (B) Differentiate into various cell types
- (C) Produce antibodies
- (D) Store genetic information only

Q.450

MCQ

The type of immunity provided by vaccines is:

- (A) Innate immunity
- (B) Passive acquired immunity
- (C) Active acquired immunity
- (D) Autoimmunity

Q.451

MCQ

B-lymphocytes produce:

- (A) T-cells
- (B) Antibodies
- (C) Antigens
- (D) Interferons

Q.452

MCQ

The primary lymphoid organ responsible for T-cell maturation is the:

- (A) Spleen
- (B) Lymph node
- (C) Thymus
- (D) Bone marrow

Q.453

MCQ

Anaphylaxis is a severe form of:

- (A) Autoimmune disease
- (B) Allergic reaction
- (C) Bacterial infection
- (D) Viral infection

Q.454

MCQ

Hemoglobin carries oxygen due to the presence of:

- (A) Calcium
- (B) Haem group with Fe^{2+}
- (C) Zinc
- (D) Copper

Q.455

MCQ

The normal pH of human blood is:

- (A) 6.8–7.0
- (B) 7.0–7.2
- (C) 7.35–7.45
- (D) 7.5–7.8

Q.456

SHORT

Insulin is produced by the _____ cells of the islets of Langerhans.

Ans: _____

Q.457

MCQ

The enzyme pepsin in the stomach digests:

- (A) Starch
- (B) Fats
- (C) Proteins
- (D) Nucleic acids

Q.458

SHORT

Peristalsis is the movement of the _____ in the alimentary canal.

Ans: _____

Q.459

SHORT

Bowman's capsule is the site of _____ in the kidney.

Ans: _____

Q.460

MCQ

The process of producing urine from blood involves filtration, reabsorption, and:

- (A) Digestion
- (B) Secretion
- (C) Excretion only
- (D) Osmosis only

Q.461

MCQ

The primary function of the large intestine is:

- (A) Digestion of proteins
- (B) Absorption of water and mineral salts
- (C) Secretion of bile
- (D) Production of enzymes

Q.462

SHORT

What is the sliding filament theory of muscle contraction?

Ans: _____

Q.463

SHORT

Acetylcholine is a _____ at the neuromuscular junction.

Ans: _____

■ Plant Physiology

Q.464

SHORT

Guard cells regulate _____ in plants.

Ans: _____

Q.465

MCQ

The hormone that promotes fruit ripening is:

- (A) Auxin
- (B) Cytokinin
- (C) Ethylene
- (D) ABA

Q.466

MCQ

Abscisic acid (ABA) promotes:

- (A) Stem elongation
- (B) Seed germination
- (C) Stomata closure and seed dormancy
- (D) Root growth

Q.467

MCQ

The movement of sugar from source to sink through phloem is called:

- (A) Transpiration
- (B) Translocation
- (C) Osmosis
- (D) Active transport

Q.468

MCQ

Nitrogen fixation is carried out by which organisms?

- (A) All bacteria
- (B) Only Rhizobium
- (C) Cyanobacteria and Rhizobium
- (D) Fungi

Q.469

MCQ

Which pigment absorbs red and far-red light in plants?

- (A) Chlorophyll
- (B) Carotenoid
- (C) Phytochrome
- (D) Anthocyanin

Q.470

SHORT

Define photoperiodism and explain its significance in agriculture.

Ans: _____

Q.471

SHORT

What is vernalization? Which crops benefit from it?

Ans: _____

■ Ecology

Q.472

MCQ

Succession leading to a stable climax community starting from bare rock is called:

- (A) Hydrarch succession
- (B) Xerarch succession
- (C) Primary succession
- (D) Secondary succession

Q.473

MCQ

Eutrophication is caused by excessive:

- (A) CO₂ in air
- (B) Nutrient enrichment of water bodies
- (C) UV radiation
- (D) Soil erosion

Q.474

MCQ

The ozone hole is primarily caused by:

- (A) CO₂
- (B) Methane
- (C) CFCs (chlorofluorocarbons)
- (D) Nitrous oxide

Q.475

MCQ

BOD (Biological Oxygen Demand) is a measure of:

- (A) Oxygen in atmosphere
- (B) Organic pollution in water
- (C) Biodiversity
- (D) Soil fertility

Q.476

MCQ

An organism that obtains energy by breaking down dead organic matter is called a:

- (A) Producer
- (B) Consumer
- (C) Decomposer
- (D) Parasite

Q.477

MCQ

Biomagnification refers to the increase in concentration of toxins with:

- (A) Decreasing temperature
- (B) Increasing trophic levels
- (C) Increasing biodiversity
- (D) Decreasing population

Q.478

MCQ

Ex-situ conservation includes:

- (A) National parks
- (B) Biosphere reserves
- (C) Zoological parks and seed banks
- (D) Wildlife sanctuaries

Q.479

MCQ

Project Tiger in India was launched in:

- (A) 1965
- (B) 1973
- (C) 1982
- (D) 1991

■ Reproduction

Q.480

MCQ

Which type of reproduction leads to greater genetic variation?

- (A) Binary fission
- (B) Budding
- (C) Sexual reproduction
- (D) Parthenogenesis

Q.481

MCQ

Placenta in mammals is the site of exchange of:

- (A) Oxygen and CO₂ only
- (B) Nutrients, gases, and waste between mother and foetus
- (C) Blood cells
- (D) Hormones only

Q.482

MCQ

Hormone responsible for parturition (childbirth) is:

- (A) Progesterone
- (B) Estrogen
- (C) Oxytocin
- (D) Prolactin

Q.483

MCQ

The corpus luteum secretes:

- (A) FSH
- (B) LH
- (C) Progesterone
- (D) Testosterone

■ Genetics

Q.484

MCQ

In humans, sex determination is based on:

- (A) Temperature
- (B) XY chromosomal system
- (C) XX chromosomal system
- (D) Environmental factors

Q.485

MCQ

Down syndrome (Trisomy 21) results from:

- (A) Deletion of chromosome 21
- (B) Monosomy of chromosome 21
- (C) Non-disjunction leading to extra chromosome 21
- (D) Point mutation

Q.486

SHORT

Sickle cell anaemia is caused by a mutation in the _____ chain of haemoglobin.

Ans: _____

Q.487

MCQ

Turner syndrome (XO) patients are:

- (A) Male, infertile
- (B) Female, infertile
- (C) Male, fertile
- (D) Female, fertile

■ Cell Biology

Q.488

MCQ

The total number of chromosomes in a human somatic cell is:

- (A) 23
- (B) 46
- (C) 48
- (D) 22

■ Plant Physiology

Q.489

MCQ

The source of oxygen released in photosynthesis is:

- (A) CO₂
- (B) Glucose
- (C) Water (H₂O)
- (D) ATP

Q.490

MCQ

Photophosphorylation is the synthesis of ATP using energy from:

- (A) Respiration
- (B) Light
- (C) Chemical reactions
- (D) Fermentation

Q.491

MCQ

Ferredoxin functions as an electron carrier during:

- (A) Dark reaction
- (B) Light reaction in PSI
- (C) Glycolysis
- (D) Krebs cycle

Q.492

MCQ

CAM plants open their stomata at night to:

- (A) Lose water
- (B) Fix CO₂ as malic acid
- (C) Absorb oxygen
- (D) Release excess sugar

■ Reproduction

Q.493

MCQ

Apomixis is the formation of seeds without:

- (A) Water
- (B) Fertilization
- (C) Sunlight
- (D) Soil nutrients

Q.494

SHORT

A polyembryonic seed contains _____ embryo/embryos.

Ans: _____

Q.495

SHORT

Vivipary is the phenomenon of seeds germinating _____ the fruit.

Ans: _____

■ Biotechnology

Q.496

SHORT

Transgenic animals are those that have a _____ gene introduced into their genome.

Ans: _____

Q.497

MCQ

DNA fingerprinting uses:

- (A) PCR only
- (B) Satellite DNA (VNTRs)
- (C) Gel electrophoresis only
- (D) Restriction enzymes only

Q.498

SHORT

The Human Genome Project revealed that the human genome has approximately _____ protein-coding genes.

Ans: _____

Q.499

MCQ

RNAi (RNA interference) is a natural defense mechanism against:

- (A) Bacteria
- (B) Foreign dsRNA/viruses
- (C) Fungi
- (D) UV radiation

Q.500

MCQ

Totipotency is the capacity of a cell to develop into:

- (A) Any organ
- (B) A complete organism
- (C) A specific tissue only
- (D) A gamete

■ Ecology

Q.501

MCQ

The pioneer species in a xerarch succession is:

- (A) Mosses
- (B) Lichens
- (C) Grasses
- (D) Trees

Q.502

MCQ

An ecological pyramid of energy is always:

- (A) Inverted
- (B) Spindle-shaped
- (C) Upright
- (D) Irregular

Q.503

SHORT

Niche of an organism is its _____ role in an ecosystem.

Ans: _____

Q.504

MCQ

The Convention on Biological Diversity (CBD) was signed at:

- (A) Stockholm, 1972
- (B) Rio de Janeiro, 1992
- (C) Kyoto, 1997
- (D) Paris, 2015

Q.505

SHORT

Why is the Amazon rainforest called the 'lungs of the Earth'?

Ans: _____

■ Evolution

Q.506

MCQ

The fossil evidence for evolution is found in:

- (A) Museums only
- (B) Sedimentary rocks
- (C) Igneous rocks
- (D) Metamorphic rocks

Q.507

MCQ

Industrial melanism in *Biston betularia* moths is an example of:

- (A) Mutation
- (B) Natural selection in action
- (C) Genetic drift
- (D) Sexual selection

Q.508

MCQ

The theory of chemical evolution was proposed by:

- (A) Darwin
- (B) Haldane and Oparin
- (C) Lamarck
- (D) De Vries

Q.509

MCQ

Vestigial organs support the theory of evolution because:

- (A) They have no function
- (B) They represent organs functional in ancestors
- (C) They are present in all animals
- (D) They improve fitness

Q.510

SHORT

The appendix in humans is considered a _____ organ.

Ans: _____

■ Human Physiology

Q.511

MCQ

The most abundant immunoglobulin in serum is:

- (A) IgM
- (B) IgA
- (C) IgG
- (D) IgE

Q.512

MCQ

Active immunity differs from passive immunity in that active immunity:

- (A) Provides faster protection
- (B) Is acquired by antibody transfer
- (C) Involves self-produced antibodies
- (D) Is always innate

Q.513

SHORT

Pneumonia is caused by _____ and affects the _____ of lungs.

Ans: _____

Q.514

SHORT

Explain the role of the hypothalamus in body temperature regulation.

Ans: _____

Q.515

MCQ

The resting membrane potential of a neuron is approximately:

- (A) +70 mV
- (B) 0 mV
- (C) -70 mV
- (D) -35 mV

Q.516

SHORT

Schwann cells form the _____ sheath around nerve fibres.

Ans: _____

Q.517

MCQ

Which lobe of the brain is responsible for vision?

- (A) Frontal lobe
- (B) Temporal lobe
- (C) Parietal lobe
- (D) Occipital lobe

Q.518

MCQ

Wernicke's area is responsible for:

- (A) Motor speech
- (B) Understanding language
- (C) Vision
- (D) Balance

Q.519

SHORT

Mention two differences between rod cells and cone cells in the retina.

Ans: _____

ANSWER KEY — BIOLOGY

Q.377: C	Q.378: B	Q.379: B	Q.380: B
Q.381: C	Q.382: B	Q.383: B	Q.386: B
Q.387: B	Q.388: C	Q.389: C	Q.391: A
Q.392: B	Q.394: B	Q.395: B	Q.396: B
Q.397: B	Q.398: B	Q.399: B	Q.400: D
Q.401: A	Q.402: B	Q.403: B	Q.407: B
Q.409: B	Q.410: B	Q.411: B	Q.412: B
Q.414: B	Q.415: B	Q.416: C	Q.417: B
Q.419: B	Q.421: B	Q.422: B	Q.423: C
Q.425: B	Q.426: B	Q.427: C	Q.429: B
Q.430: C	Q.431: C	Q.432: B	Q.434: C
Q.436: C	Q.437: C	Q.439: B	Q.440: C
Q.442: B	Q.443: C	Q.444: D	Q.445: C
Q.446: B	Q.447: B	Q.448: B	Q.449: B
Q.450: C	Q.451: B	Q.452: C	Q.453: B
Q.454: B	Q.455: C	Q.457: C	Q.460: B
Q.461: B	Q.465: C	Q.466: C	Q.467: B
Q.468: C	Q.469: C	Q.472: C	Q.473: B
Q.474: C	Q.475: B	Q.476: C	Q.477: B
Q.478: C	Q.479: B	Q.480: C	Q.481: B
Q.482: C	Q.483: C	Q.484: B	Q.485: C
Q.487: B	Q.488: B	Q.489: C	Q.490: B
Q.491: B	Q.492: B	Q.493: B	Q.497: B
Q.499: B	Q.500: B	Q.501: B	Q.502: C
Q.504: B	Q.506: B	Q.507: B	Q.508: B
Q.509: B	Q.511: C	Q.512: C	Q.515: C
Q.517: D	Q.518: B		

Note: For SHORT ANSWER and NUMERICAL questions, refer to standard textbooks for detailed solutions.